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II/IV-B.Tech(EVEN Sem), Academic Year: 2025-2026
B.Tech(CSE, CSIT &ME), 2024 Batch II/IV, EVEN Semester
Subject Code: 24MT2019 / TITLE: Probability and Statistics
Home Assignment-4 (CO-4)

1. Ten individuals have participated in a diet-modification program to stimulate weight loss. Their weight both before and after participation in the program is shown in the following list.

Subject	1	2	3	4	5	6	7	8	9	10
Before	195	213	247	201	187	210	215	246	294	310
After	187	195	221	190	175	197	199	221	278	285

Is there evidence to support the claim that this particular diet-modification program is effective in producing a mean weight reduction? Use α = 0.05.

2. A product developer is interested in reducing the drying time of a primer paint. Two formulations of the paint are tested; formulation 1 is the standard chemistry, and formulation 2 has a new drying ingredient that should reduce the drying time. From experience, it is known that the standard deviation of drying time is 8 minutes, and this inherent variability should be unaffected by the addition of the new ingredient. Ten specimens are painted with formulation 1, and another 10 specimens are painted with formulation 2; the 20 specimens are painted in random order. The two sample average drying times are $\bar{x}_1 = 121$ minutes and $\bar{x}_2 = 112$ minutes, respectively. What conclusions can the product developer draw about the effectiveness of the new ingredient, using α = 0.05? Apply appropriate formula for test of hypothesis.

3. 1000 students at college level were graded according to their I.Q. and the economic conditions of their home. Use Chi-Square test to find out whether there is any association between condition at home and I.Q.(Use α-value of 0.05).

Economic Condition	High	Low	**Total**
Rich	460	140	**600**
Poor	240	160	**400**
Total	**700**	**300**	**1000**

4. A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be : 12.1, 11.9, 12.4, 12.3, 11.5, 11.6, 12.1 and 12.4. Obtain 95% confidence limits for the mean weight of the envelopes received at that post office.

5. Three samples obtained from normal poulations with equal variances. Test the hypothesis that sample means are equal.

8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

6. The means of two single large samples of 1,000 and 2,000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of standard deviation 2.5 inches? Test at 5% level of significance

7. Two types of new cars produced in India are tested for petrol mileage, one sample is consisting of 42 cars averaged 15 Kmpl while the other sample consisting of 80 cars averaged 11.5 Kmpl with the population variances 2.0 and 1.5 respectively. Test whether there is any significance difference in the petrol consumption of these two types of cars. Use 1% level of confidence.

8. A researcher wanted to analyze whether the mean housing prices are the same regardless of the three crowded places where they are located. A random samples of the prices at which the houses are being purchased at all the three areas are given below:

Observation	Low	Moderate	High
1	100	71	39
2	62	59	66
3	32	110	34
4	102	65	54
5	63	90	46

Calculate F and report your results with the α-value of 0.01.

9. A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kilograms with a standard deviation of 0.5 kilograms. The company wishes to test the hypothesis H_0:$\mu = 12$ against H_1: $\mu < 12$, using a random sample of four specimens. Use $\alpha = 5\%$. Apply t-Test for single mean formula for test of hypothesis.

10. Medical researchers have developed a new artificial heart constructed primarily of titanium and plastic. The heart will last and operate almost indefinitely once it is implanted in the patient's body, but the battery pack needs to be recharged about every four hours. A random sample of 50 battery packs is selected and subjected to a life test. The average life of these batteries is 4.05 hours. Assume that battery life is normally distributed with standard deviation $\sigma = 0.2$ hour. Is there evidence to support the claim that mean battery life exceeds 4 hours? Use $\alpha = 0.05$.